

AI hedging: baselines versus the simple-profile AI run

Current SPY experiment | Simple jump stresses | Re-run results, 10 October 2026

What changed in this version. Both runs were repeated. The terminal-loss limit is now **\$50** (previous version: \$100) with the same \$1,000 capital, calibration, seeds and path counts. The AI was re-run (50 proposals); the baselines were re-run against the same saved setup. All numbers in Sections 8–10 come from `report-simple.json` (AI) and `report-baselines.json` (baselines). I independently rebuilt every scenario set from the project code and reproduced the reported metrics exactly (31/31 checks, relative tolerance 10^{-9}).

1. The question we are testing

Can AI find a stock-hedging rule that meets portfolio risk limits and costs less to execute than simple rules or a numerical search? The experiment holds one short European call on one share, funded by investor capital. It is not a stock-price forecasting competition.

Choose policy (b_0, k) to minimize mean base trading fees, subject to every scenario group $g \in \{\text{base, more_jumps}\}$ passing:

$$\text{CVaR}_{0.95}(\max \text{drawdown}_g) \leq 10\%, \quad \text{CVaR}_{0.95}(\text{terminal loss}_g) \leq \$50, \quad \min \text{cash}_g \geq \$0, \quad \min \text{equity}_g > \$0.$$

The comparison uses \$1,000 capital. These constraints are acceptance assumptions. Tightening the loss limit from \$100 to \$50 creates a different experiment: results in this version are not directly comparable with the previous score-card.

2. End-to-end workflow

Stage	What happens
1. Historical input	Fetch about five years of daily SPY prices, or load a CSV. Record the data period and source.
2. Fit a market model	Estimate ordinary volatility, drift, and rough jump frequency/size from unusually large returns.
3. Freeze search scenarios	Generate 2,000 possible years per scenario group, each with 252 daily steps. Reuse these exact paths for all candidates.
4. Propose and evaluate	AI proposes two parameters. The simulator executes trades; the evaluator returns cost and constraint violations.
5. Select a policy	Among passing candidates, select the lowest base-fee policy. If none pass, report no feasible candidate.
6. Independent evaluation	Run the selected policy on 2,000 fresh paths per group. Compare with baselines under the same setup.

Each path has 253 price observations: the initial price and 252 subsequent prices. The simple profile has two groups, so one candidate receives 4,000 search path evaluations.

3. Market generation and simple jump stress

Historical SPY prices inform a synthetic model; they are not replayed as the test paths. Each day combines a drift term, normal random fluctuations, and random jumps. The number of jumps follows a Poisson distribution; each jump has a normally distributed log size, so its price multiplier is lognormal.

$$N \sim \text{Poisson}(\lambda \Delta t), \quad Y \sim \mathcal{N}(m, s^2), \quad \kappa = e^{m+s^2/2} - 1, \\ \ln \frac{S_{t+\Delta t}}{S_t} = \left(\mu - \lambda \kappa - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z + \sum_{j=1}^N Y_j, \quad Z \sim \mathcal{N}(0, 1).$$

μ is the fitted expected return, σ is ordinary volatility, λ is annual jump frequency, and Z is standard normal. The compensator $\lambda\kappa$ preserves the chosen expected return. Jumps have random timing; no special step-63 event exists.

Group	Definition
Base	Use fitted frequency and jump-size distribution ($\lambda = 8.44/\text{year}$).
More jumps	Increase fitted frequency by 25% ($\lambda = 10.55/\text{year}$); keep jump-size mean and variability unchanged.

Realised in the generated paths: 8.49 and 10.64 jumps per path (search), 8.40 and 10.42 (fresh). The additional 25% is an induced sensitivity test. All simulated events are synthetic. The 42 extreme historical days used in fitting are proxies, not confirmed discrete jump events. The simple profile omits earlier large jump-size shifts and the separate high-volatility group.

4. What the hedge does every day

$$E = \text{cash} + \text{shares} \cdot S - \text{option value}, \quad D = \frac{\text{running peak } E - E}{\text{running peak } E}, \quad \text{band} = \frac{b_0}{1 + kD}.$$

If $|\text{target delta} - \text{shares}| > \text{band}$: rebalance to delta.

b_0 is tolerance measured in shares; k controls tightening during drawdowns. The simulator accrues interest, charges fees on trades, tracks cash/equity before and after trading, and liquidates the hedge and settles the option at expiry.

Black–Scholes calculates the option value and target delta using current spot, fixed strike, remaining time, rate and fixed marking volatility. It does not generate stock prices. Calibration currently sets strike equal to the latest spot (\$778.57), creating an at-the-money benchmark; strike then remains fixed.

5. AI versus numerical calculation

The AI selects parameters. Python calculates the outcomes and enforces acceptance rules. Compact memory retains best/recent candidates. Model weights are not trained. Duplicate or malformed proposals consume model calls but do not trigger a new policy simulation. In this run the proposer was Qwen3-8B (temperature 0.6), served through an OpenAI-compatible endpoint.

6. Metrics to show in a comparison

Metric	Meaning and evaluation rule
Fresh-test feasibility	Primary gate: every group must meet all risk limits. A cheap failing policy is not a winner.
Base mean fees (\$)	Average stock execution fees per simulated year in the base group. Lower is better among passing policies.
Drawdown CVaR95 (%)	Compute the largest peak-to-trough fall in each year, then average the worst 5% of those values.
Terminal-loss CVaR95 (\$)	Average the worst 5% of positive final shortfalls against capital grown at the cash rate.
Minimum cash / equity (\$)	Worst observed cash and equity over all paths and times. Cash must be nonnegative and equity positive.
Mean final equity (\$)	Shows typical financial outcome. Passing tail-risk limits does not guarantee positive returns.
Search effort	Unique simulator evaluations, model requests, duplicate/invalid rate, tokens, and measured runtime if recorded.

drawdown = (peak – current equity)/peak; terminal shortfall = $\max(\text{capital} \cdot e^{rT} - \text{final equity}, 0)$; at 95% confidence, tail = worst 5% = 100 of 2,000 paths.

Confidence selects the tail size; it is not the probability that a policy succeeds. At \$1,000 capital and 4% interest, the one-year target is \$1,040.81. CVaR limits constrain tail averages, not the largest individual loss.

7. Baselines and fair comparison

Method	Policy / budget
Daily delta hedge	$b_0 = 0, k = 0$. Rebalance each trading day.
Fixed band	$b_0 = 0.10, k = 0$. No drawdown adaptation.
Numerical grid	50 distinct combinations: $b_0 \in \{0.01, 0.03, 0.05, 0.075, 0.10, 0.15, 0.20, 0.30, 0.40, 0.50\} \times k \in \{0, 2, 5, 10, 20\}$. Select lowest-fee feasible candidate.
AI search	Proposes $b_0 \in [0, 0.5]$ and $k \in [0, 20]$ from feedback. 50 proposals; compare at the same number of unique evaluations as the grid.

Keep calibration, spot/strike, capital, fees, limits, scenario profile, seeds and path counts identical. Compare each method on the same fresh paths, without feeding their results back into policy selection. Repeatedly tuning to the same final-test seed turns it into development data; use new untouched seeds for the final presentation.

8. Re-run results: baselines versus the simple-profile AI run

All four methods use the same configuration (spot = strike = \$778.57, $T = 1$, 252 steps, $r = 4\%$, marking volatility 13.75%, fee 5 bp, capital \$1,000), the same search paths (seeds 10 and 11) and the same fresh paths (seeds 1,000,010 and 1,000,011), 2,000 paths per group. “Worst” means the larger of the two scenario groups.

Scorecard at the run limits (drawdown CVaR95 $\leq 10\%$, terminal-loss CVaR95 $\leq \$50$)

Method	(b_0, k)	Unique evals	Search fees	Fresh fees	Worst DD CVaR	Worst loss CVaR	Min cash	Fresh
AI search	(0.18, 11)	23	\$1.021	\$1.022	2.81%	\$47.81	\$201.08	PASS
Grid search	(0.20, 20)	50	\$0.985	\$0.981	2.90%	\$48.34	\$184.11	PASS
Fixed band	(0.10, 0)	1	\$1.215	\$1.210	2.55%	\$44.22	\$206.59	PASS
Daily hedge	(0, 0)	1	\$2.251	\$2.251	2.48%	\$44.62	\$180.34	PASS
Fresh base paths	AI (0.18, 11)	Grid (0.20, 20)		Fixed band		Daily		
Rebalances / year	5.0	4.4		11.5		244.1		
Mean final equity	\$1,026.90	\$1,026.32		\$1,028.14		\$1,028.67		
Min equity (worst group)	\$960.38	\$960.02		\$952.89		\$951.12		

Read across the scorecard: all four selections pass both fresh groups. The grid winner is the cheapest passing policy (\$0.981). The AI selection costs **4.1% more than the grid** (+\$0.040), but **15.6% less than the fixed band** and **54.6% less than daily hedging**. Search-to-fresh drift is small for every method (at most \$0.005, fixed band), so selection did not overfit the search paths noticeably.

Is the AI–grid gap real or Monte Carlo noise? Both policies are evaluated on the *same* 2,000 fresh base paths, so the per-path fee difference $d_i = f_i^{\text{AI}} - f_i^{\text{grid}}$ is a paired sample (common random numbers). Its mean is \$0.0404 with standard error \$0.0024 (95% CI [\$0.036, \$0.045]); per-path fees are highly correlated ($\rho = 0.92$), which shrinks the standard error about 3.6× relative to an unpaired comparison (\$0.0087). On these paths the grid is cheaper beyond sampling noise. The CI covers path noise only, not selection noise or model risk. Paired CIs versus the baselines: AI–fixed = $-\$0.188 [-0.198, -0.179]$; AI–daily = $-\$1.229 [-1.255, -1.203]$.

Which constraint binds. The terminal-loss limit is the active constraint; the drawdown limit is slack (all passing candidates sit near 2.5–3.0% versus the 10% limit). On fresh more-jumps paths the AI's loss CVaR is \$47.81 with bootstrap standard error \$1.07 (2,000 resamples), i.e. about 2.0 standard errors inside the \$50 limit; the grid's \$48.34 has SE \$1.01, about 1.6 SE inside. The fraction of bootstrap resamples breaching \$50 is 2.1% for the AI and 4.7% for the grid: the grid buys its lower fee with a thinner safety margin.

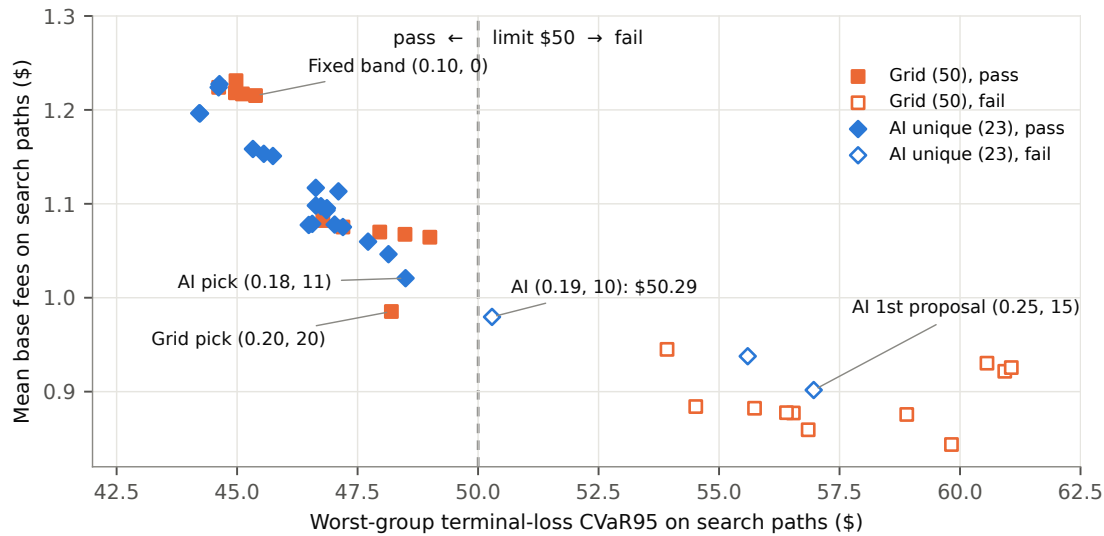


Figure 1: Cost–risk trade-off on the search paths. Each marker is one evaluated policy; filled markers pass all constraints, hollow markers fail. Fees fall as policies accept more terminal-loss risk; the \$50 limit cuts the frontier near \$0.98–1.02. View is cropped to the frontier region: grid policies with $b_0 \leq 0.075$ (fees \$1.34–\$2.13, all passing) and with loss CVaR above \$62 (all failing) lie outside the plot.

Search effort. AI: 50 model calls, 23 unique evaluations (46%), 27 duplicates (54%), 0 malformed replies, 58,630 tokens (57,434 prompt + 1,196 completion), 20 of 23 evaluated policies feasible. Grid: 50 unique evaluations, 31 of 50 feasible. The AI reached within 4.1% of the grid optimum with fewer than half the simulator evaluations, but the budgets were unequal and its search stalled: after proposal 30 the model produced no new policy. Its first proposal, (0.25, 15)—the previous version's winner under the \$100 limit—now fails (loss CVaR \$56.96). The AI never explored the corner ($b_0 \approx 0.20$, $k \approx 20$) where the grid winner lies, even though $k = 20$ was inside its allowed range; its unique proposals stayed within $b_0 \in [0.10, 0.25]$, $k \in [8, 18]$.

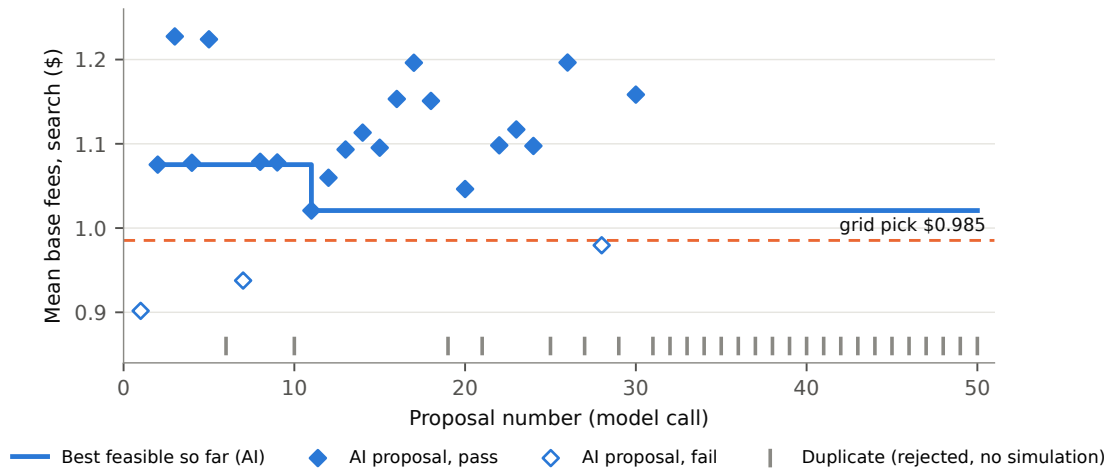


Figure 2: AI search trajectory. 23 of 50 proposals were new policies; 27 were exact duplicates and were rejected without simulation. The best feasible policy was found at proposal 11 and never improved; proposals 31–50 alternated between two already-evaluated policies, (0.15, 14) and (0.14, 13).

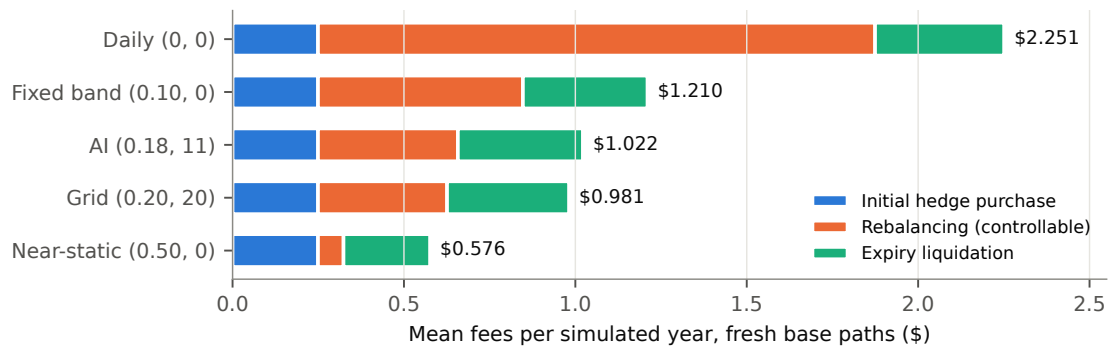


Figure 3: Where the fees come from (fresh base paths). Every policy pays \$0.249 to buy the initial hedge ($\Delta_0 = 0.640$ shares) and \$0.25–0.38 to liquidate at expiry; only the rebalancing component responds to (b_0, k) . A near-static hedge (0.50, 0), which fails the risk limits, shows the floor (\$0.576). The AI–grid gap of \$0.040 is about 8% of the AI’s controllable rebalancing fees (\$0.408 vs \$0.376) plus \$0.009 of liquidation fees.

9. AI versus baselines across risk appetites

The question: Can AI find a cheaper policy while staying within the same risk limits? Choose the risk appetite first, then compare passing policies. Capital stays at \$1,000 and cash must remain nonnegative in every tier.

Risk appetite	Drawdown CVaR95 limit	Terminal-loss CVaR95 limit
Conservative	3%	\$50
Balanced	5%	\$75
Flexible	10%	\$100
Run limits (this AI run)	10%	\$50

These tiers are illustrative choices for this experiment, not industry standards. The re-run AI search was steered by the run limits; the other columns re-select from saved search records.

Scorecard: average trading fees on fresh base paths (selected policy in brackets)

Method	Conservative	Run limits	Balanced	Flexible
AI candidates (23)	\$1.044 PASS (0.17, 14)	\$1.022 PASS (0.18, 11)	\$0.900 PASS (0.25, 15)	\$0.900 PASS (0.25, 15)
Grid search (50)	\$1.064 PASS (0.15, 2)	\$0.981 PASS (0.20, 20)	\$0.806 PASS (0.50, 20)	\$0.684 PASS (0.40, 2)
Fixed band (0.10)	\$1.210 PASS	\$1.210 PASS	\$1.210 PASS	\$1.210 PASS
Daily hedge	\$2.251 PASS	\$2.251 PASS	\$2.251 PASS	\$2.251 PASS

Read down each column: all displayed selections pass that tier on both fresh scenario groups; lower fees are better (cheapest in bold). The AI pool is cheapest only in the Conservative tier: \$1.044 versus grid \$1.064, a paired difference of $-\$0.019$ (95% CI $[-0.024, -0.014]$), i.e. 1.8% cheaper. The grid is cheaper at the run limits (4.1%), Balanced (11.7%) and Flexible (31.5%). In the looser tiers the comparison is tilted toward the grid: the AI's candidates were generated under the \$50 limit and never explore wide bands, whereas the grid covers b_0 up to 0.50 by design.

Versus the previous version: the fixed-band and daily rows are unchanged (\$1.210 and \$2.251) because calibration, seeds and fresh paths are identical. The grid column values match the previous scorecard (\$1.064, \$0.806, \$0.684) for the same reason. The AI column changed because the AI was re-run under a different loss limit; its Conservative selection now costs \$1.044 versus \$1.026 in the previous run (1.8% more expensive) but is still cheaper than the grid. In the Run-limits column the previous AI winner, (0.25, 15), is no longer admissible.

10. The evaluation rule in three steps

1. **Pass:** satisfy all limits on both fresh scenario groups.
2. **Compare:** among passing methods in the same column, lower fresh base fees are better; report the paired fee difference with its confidence interval.
3. **Check effort:** compare AI and grid at 50 unique simulator evaluations, with identical calibration and paths. Also report model calls, duplicate proposals and tokens.

Evidence note: Conservative, Balanced and Flexible reselect saved candidates using search metrics only; they are not new AI runs. The budgets were unequal (23 vs 50 unique evaluations). Repeat with equal budgets, a duplicate-avoiding proposal step and untouched seeds before claiming an advantage either way.

11. Assumptions and practical limits

This is a one-share, cash-settled European-call benchmark, not a full model of a listed SPY contract. Fixed Black–Scholes marks omit jump-aware option pricing. Historical extreme-day fitting is a heuristic; normal fluctuations and jumps are not cleanly separated. Adjusted returns proxy total returns, while explicit dividends, early exercise, margin calls, market impact and volatility clustering are omitted. Cash interest is symmetric and trading fees are proportional.

Fresh synthetic paths test robustness under the assumed model. A separate historical out-of-time backtest and better valuation/transaction modeling are needed to assess real-market performance. Reducing stress intensity changes the test; it does not itself improve the hedge.

12. Files and results

`spy_calibration.py` → historical fit; `market_jumps.py` → paths; `portfolio.py` → hedge execution; `risk.py` → metrics; `portfolio_agent_realworld.py` → AI loop; `baseline_spy.py` → comparison. AI JSON: best, final_test, records, attempts, token_usage. Baseline JSON: methods, selected, final_test. Keep full configuration and calibration metadata with every result.

Sources: AI run in `report-simple.json` (calibration window 11 Oct 2021 – 9 Oct 2026, 1,255 daily observations, 42 extreme days); baselines in `report-baselines.json` (generated from the same saved AI setup). Paired confidence intervals, bootstrap CVaR errors, rebalance counts, fee decomposition and tier re-selection were computed for this report by re-running the project's simulator on the identical seeds. This version supersedes the previous \$100-limit scorecard.

Glossary: parameters, metrics and model terms

Use this page as a key to the symbols and short names used throughout the guide.

Term	Plain-language definition
Base band (b_0)	The normal allowed gap between target option delta and shares held. Measured in shares. A wider band usually means fewer trades and lower fees, with more hedge mismatch.
Drawdown response (k)	How much the band tightens after a fall from the portfolio peak. In $\text{band} = b_0 / (1 + kD)$, larger k makes the band shrink faster as drawdown D increases.
Band (b)	The current share-gap tolerance. If $ \text{target delta} - \text{shares held} $ exceeds this value, the policy rebalances.
Policy / configuration	The rule and its numeric choices. Here the AI policy is the pair (b_0, k) ; market configuration includes capital, spot, strike, horizon and rates.
Spot (S, S_0)	Current price of one share. S_0 is its starting price. Spot changes along a simulated path.
Strike (K)	Fixed price written into the call contract. It does not change as spot moves. At-the-money means initial spot equals strike.
Call option	Contract giving its buyer the right to buy the share at strike K by expiry. The seller has the matching obligation.
Short call / liability	The portfolio has sold a call, collected premium and may owe its payoff. For this benchmark it settles in cash at expiry.
Maturity / horizon (T)	Time until the option expires. One year here. Remaining time falls toward zero each daily step.
Step / Δt	One time interval in a simulated year. With 252 steps/year, each step is roughly one trading day.
Path / scenario	One possible sequence of stock prices over the whole year. 2,000 paths means 2,000 possible years, not 2,000 days.
Seed	Number that initializes random generation. Reusing a seed reproduces paths; changing it generates new paths.
Expected return (μ)	Fitted average annual price-return assumption in the jump model. It is not a promise or forecast.
Diffusion volatility (σ)	Annualized size of ordinary random price moves, excluding the modeled jump component.
Jump intensity (λ)	Expected number of modeled jump events per year. Actual count varies randomly.
Poisson distribution	Random count model used for how many jumps occur during an interval. It allows zero, one or multiple events.
Jump log size (Y)	Random change to log price at an event. Y is modeled as normal; e^Y is the event's price multiplier.
Jump mean (m) and std (s)	Average and spread of modeled jump log sizes. These describe the fitted event-size distribution.
Normal distribution / Z	Bell-shaped random-number distribution used for ordinary daily movement and jump-size variation.
Compensator ($\lambda\kappa$)	Drift adjustment for the average effect of jumps, so adding jumps does not accidentally change the specified expected return.
Calibration	Estimating model inputs from historical SPY prices. In this code, unusually large daily returns are only proxies for jumps.
Historical vs synthetic	Historical prices are used to estimate inputs. Simulated test paths are newly generated by the model; they are not historical replay.
Induced stress	An intentionally altered simulation assumption, such as increasing jump frequency by 25%. It is a sensitivity test, not an observed market event.
Black–Scholes mark	Estimated option value computed from current spot, strike, remaining time, interest rate and fixed volatility. It does not forecast the next spot price.
Delta	Estimated change in option value for a \$1 move in spot. It also serves as the hedge's target number of shares per option in this simplified setup.

Term	Plain-language definition
Cash / equity	Cash is uninvested or borrowed balance. $\text{Equity} = \text{cash} + \text{shares} \cdot \text{spot} - \text{current option liability}$.
Peak equity	Highest recorded portfolio equity so far. Drawdown compares current equity with this peak.
Drawdown (D)	Fractional fall from the prior equity peak: $(\text{peak} - \text{current})/\text{peak}$. A 10% fall from \$1,000 is \$100.
Maximum drawdown (MDD)	Largest peak-to-trough percentage fall during one simulated path/year.
Terminal loss / shortfall	Positive difference, if any, between the interest-grown capital target and final equity. It is zero if final equity meets or beats the target.
VaR95 / 95th percentile	Loss threshold that 95% of outcomes do not exceed; approximately the boundary of the worst 5%.
CVaR / CVaR95	Average loss in the worst 5% of outcomes at 95% confidence. The code applies this to path maximum drawdowns and terminal shortfalls.
Confidence level	Sets the tail fraction for CVaR. At 95%, use the worst 5%. It is not the probability the policy passes.
Limit / constraint	Required risk boundary. A candidate is feasible only if it satisfies every limit in every tested group.
Feasible / pass	Candidate meets all configured constraints on the sampled paths. Passing samples is not a guarantee for every possible market.
Base mean fees	Average simulated stock-trading fees in the base group. Minimized only among candidates that pass the constraints.
Normalized violation	Unitless guidance score for failed candidates. Lower means closer to limits; a score above zero is still a failure.
Fresh test / holdout	New random paths reserved to evaluate the selected rule after search. They test the assumed model, not unseen real markets.
Baseline	Simple comparison policy such as daily rebalancing or a fixed band. It shows what the AI result should be compared against.
Numerical grid	Preselected list of policy parameter pairs evaluated systematically. Useful benchmark for whether AI search adds value.
Unique simulator evaluation	One new policy evaluated on the market paths. Duplicate or invalid model responses are calls, but not unique evaluations.
Evaluation budget	Maximum number of AI proposals or simulator evaluations. Report both because invalid and duplicate proposals can waste AI calls.
AI search	Model proposes policy parameters and uses measured simulator feedback to guide later proposals. The model is not retrained.
Fresh base fees	Fees for the selected policy on new base paths. Compare only after the policy also passes risk constraints.
Paired difference	Difference between two policies' outcomes on the same simulated path, averaged over paths. Sharing paths cancels common noise and narrows the confidence interval.
Bootstrap standard error	Spread of a statistic (e.g. CVaR) when it is recomputed on paths resampled with replacement; a measure of its sampling uncertainty.
Interest / cash rate (r)	Annual rate used to grow cash and define the capital target at expiry. The model assumes the same rate for positive and negative cash.
Marking volatility	Fixed volatility input used in the Black–Scholes option value and delta. It does not change day to day in this benchmark.
Realized volatility	Volatility used to generate ordinary random market movements. It can differ from marking volatility.
Execution fee / fee rate	Charge on a stock trade. Fee equals fee rate times share price times absolute shares traded.
At-the-money (ATM)	An option whose strike equals the current share price. In this setup strike is set equal to initial spot.
Base scenario	Synthetic paths using the fitted central market parameters. It is the group used to measure and minimize average fees.

Term	Plain-language definition
Stress scenario	Synthetic paths with a deliberately changed assumption, such as 25% more jump events, to test sensitivity.
Independent paths	Fresh random paths created after policy selection. They are not used to choose the policy.
Portfolio target	Starting capital grown at the cash rate to expiry: $\text{capital} \cdot e^{rT}$. Terminal shortfall is measured from this amount.